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Chapter 12

Troubleshooting Network Communication

Knowing how to configure a Cisco router and switch is only a small portion of what you need to know in order to pass both the ICND1 and the ICND2 exams. A big part of each exam is ensuring you know how to troubleshoot problems with communication and the configuration. This chapter presents a number of common troubleshooting scenarios testing your knowledge of different commands and your ability to identify problems.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Using commands to help troubleshoot and diagnose network related problems
- Troubleshooting connectivity issues
- Using show commands to troubleshoot configuration issues
- Using debug commands to diagnose a problem

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Troubleshoot layer-1 and layer-2 characteristics of the network first when you're having connectivity issues. When troubleshooting layer 1, ensure that you are using the correct cables and that they are properly connected to the ports by checking for a link light. When troubleshooting layer 2, ensure that the data link protocol is set (Ethernet, PPP, or HDLC) and verify related settings, such as the clock rate.
- After verifying the physical aspect of the network, you can then check the logical aspect, such as the network layer (layer 3). When troubleshooting layer 3, verify the IP address configuration and routing table information.
- Ensure you know the different show commands, such as `show ip route`, `show interfaces`, `show ip interface brief`, and `show cdp neighbors detail`. Many ICND1 exam questions test your knowledge about how to troubleshoot using the `show` commands.

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Troubleshooting Commands

428–432 Choose the best answer(s).

428. You are troubleshooting communication to another device. What is the result of the command being executed in the figure below?

```
NY-R1>ping 24.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 24.0.0.2, timeout is 2 seconds:
!!!!!
```

(A) The pings were unsuccessful.

(B) The pings were successful.

(C) The traceroute was unsuccessful.

(D) The ARP cache is empty.

429. What are the results of the command executing in the figure below?

```
NY-R1>ping 24.0.0.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 24.0.0.5, timeout is 2 seconds:
.....
```

(A) The pings were unsuccessful.

(B) The pings were successful.

(C) The traceroute was unsuccessful.

(D) The ARP cache is empty.

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430. You are troubleshooting communication problems on a network on the other side of the country. What command do you use to identify at what point in the communication pathway there is a failure?

(A) `show ip arp`

(B) `ping ip_address`

(C) `telnet ip_address`

(D) `traceroute ip_address`

431. What command would you use to test VTY port configuration?

(A) `traceroute`

(B) `nslookup`

(C) `ping`

(D) `telnet`

432. You are testing connectivity to a system on a remote network. Using the output shown in the figure below, what is the default gateway setting of this workstation?

```
C:\>tracert 25.0.0.10

Tracing route to 25.0.0.10 over a maximum of 30 hops:

  1   13 ms     9 ms     7 ms     23.0.0.1
  2   13 ms    13 ms     8 ms     24.0.0.2
  3   19 ms    18 ms    23 ms     25.0.0.10

Trace complete.
```

(A) 23.0.0.1

(B) 24.0.0.2

(C) 25.0.0.10

(D) 25.0.0.1

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Troubleshooting Connectivity Issues

433–444 Choose the best answer(s).

433. You are connecting a switch to a router. What type of cable should you use?

(A) Straight-through

(B) Console

(C) Serial

(D) Crossover

434. You have your workstation connected to port 6 on the switch, but are unable to communicate on the network. What should you do first?

(A) Use `ipconfig`.

(B) Ping another system.

(C) Check the routing table.

(D) Verify your cable is connected properly.

435. You are connecting two switches together. What type of cable should you use?

(A) Straight-through

(B) Console

(C) Serial

(D) Crossover

436. You receive the following output when looking at the show interfaces command. What should you check?

```
Serial0/2/0 is down, line protocol is down
```

(A) Check the physical aspects of the link, such as the cable.

(B) Check that the IP address is set.

(C) Check that the data link protocol is configured.

(D) Verify that a route is defined to the remote network.

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437. You are the network administrator for a small network running three VLANs on a switch. You have connected a workstation to port 8 on the switch and assigned it a valid IP address for that network. You try to communicate with another system on the network, but are unable to. What could be the problem?

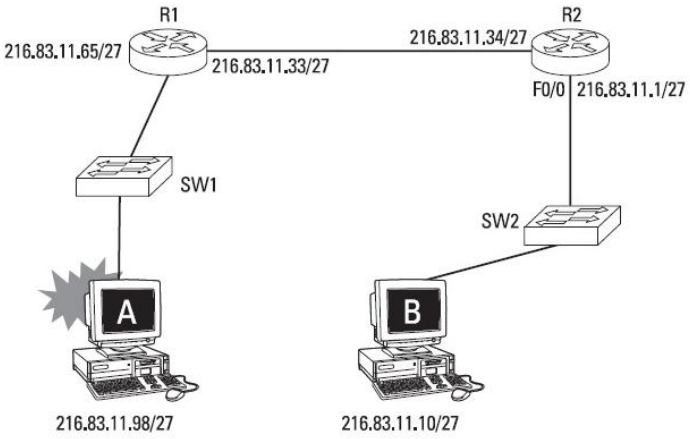
(A) You have the wrong IP assigned.

(B) CDP is running on the switch.

(C) STP is blocking communication between the VLANs.

(D) You are connected to a port in a different VLAN than the other system.

438. Looking at the figure below, Computer A is initiating communication to Computer B and sends out an ARP request. Which device responds with the ARP reply message?



(A) SW1

(B) R1

(C) R2

(D) Computer B

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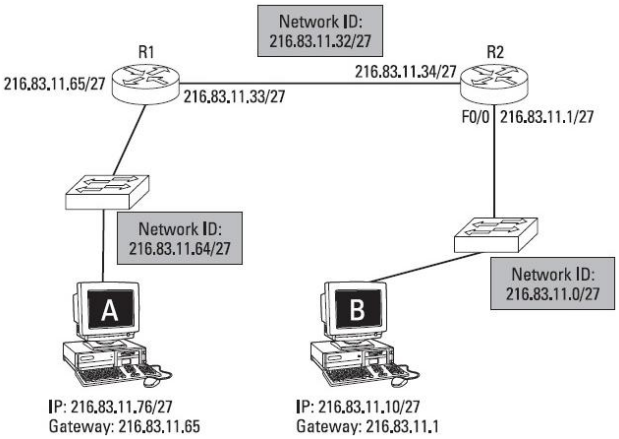
439. You are the network administrator for a company that has users located in different offices. The network configuration is shown in the figure below. You have connected the switches to the routers with a crossover cable, and the computers to the switches with straight-through cables. Computer A cannot ping Computer B. What should you do?



(A) Change the IP address of Computer A.

(B) Use crossover cables to connect computers to switches.

(C) Change the default gateway of Computer A.

(D) Use straight-through cables to connect switches to routers.

440. Having a green link light indicates that which of the following conditions have been met? (Select two.)

(A) A layer-3 protocol has been assigned at either end of the link.

(B) The network media is attached at both ends of the link.

(C) The IP address on the interface has been configured.

(D) A layer-2 protocol has been loaded and configured at either end of the link.

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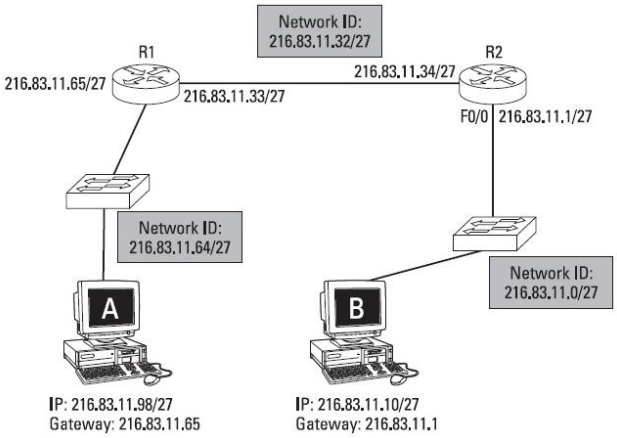
441. You are the network administrator for a company that has users located in different offices. The network configuration is shown in the figure below. You have connected the switches to the routers with a straight-through cable, and the computers to the switches with straight-through cables. Computer A cannot ping Computer B. What should you do?



(A) Change the cable type connecting the switches to the routers.

(B) Change the cable type connecting the computers to the switches.

(C) Change the IP address on Computer A.

(D) Change the IP address of Computer B.

(E) Change the IP address on R1.

442. You use the following commands to create an access control list on the FastEthernet port of the router. Users are now reporting that they cannot communicate with the intranet site. What is the problem?

```
R1(config)# access-list 102 permit tcp any any eq 23
R1(config)# access-list 102 permit tcp any any eq 25
R1(config)# interface fastethernet0/1
R1(config-if)# ip access-group 102 in
```

(A) The first line should not be permit.

(B) The second line should not be permit.

(C) There is an implied deny all traffic at the end of the ACL.

(D) The `ip access group` command should not be used.

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443. You have a switch using multiple VLANs that is connected to a router using a straight-through cable. You have configured the router as a router on the stick. You have configured the sub-interfaces on the router, but communication is not occurring between the VLANs. How do you fix the problem?

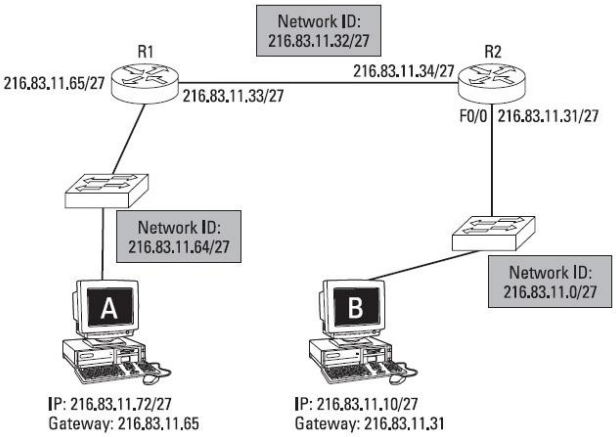
(A) Use a crossover cable to connect the switch to the router on a stick.

(B) Enable trunking on the port connecting the switch to the router.

(C) Add static routes so the router can route between the VLANs.

(D) Load RIPv2 so that routing between the VLANs can occur.

444. You are the network administrator for a company that has users located in different offices. The network configuration is shown in the figure below. You have connected the switches to the routers with straight-through cables, and the computers to the switches with straight-through cables. Computer A cannot ping Computer B. Why not?



(A) Computer B's default gateway is invalid.

(B) Computer A's default gateway is invalid.

(C) A crossover was used to connect switches to routers.

(D) Computer B's IP address is incorrect.

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Using Show Commands to Troubleshoot

445–462 Choose the best answer(s).

445. Looking at the output of the show interfaces command seen in the figure below, what is the MAC of the interface using the IP address 23.0.0.1?

```
NY-R1>show interfaces

FastEthernet0/0 is up, line protocol is up
  Hardware is Lance, address is 0010.11d9.d001 (bia 0010.11d9.d001)
  Internet address is 23.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,

 (output cut for briefness)

FastEthernet0/1 is administratively down, line protocol is down
  Hardware is Lance, address is 0010.11d9.d002 (bia 0010.11d9.d002)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set

  (Output has been cut for briefness)

Serial0/2/0 is up, line protocol is up
  Hardware is HD64570
  Internet address is 24.0.0.1/8
  MTU 1500 bytes, BW 128 Kbit, DLY 20000 usec, rely 255/255, load 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  (Output has been cut for briefness)
```

(A) HD64570

(B) ffff.ffff.ffff

(C) 0010.11d9.d002

(D) 0010.11d9.d001

446. What command was used to create the output shown in the figure below?

```
Address          Age (min)  Hardware Addr   Interface
23.0.0.1               -    0010.11D9.D001  FastEthernet0/0
23.0.0.10              4    0011.9520.8c27  FastEthernet0/0
23.0.0.100            10    0006.d6ab.a040  FastEthernet0/0

(Columns have been removed for briefness)
```

(A) `show interfaces`

(B) `show ip arp`

(C) `show ip interface brief`

(D) `show controllers`

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447. Which of the following commands can you use to figure out the IP address of a neighboring router?

(A) `show cdp neighbors`

(B) `show cdp neighbors detail`

(C) `show cdp`

(D) `cdp run`

448. Looking at the output of the show interfaces command seen in the figure below, what is the encapsulation protocol being used by the serial port?

```
NY-R1>show interfaces

FastEthernet0/0 is up, line protocol is up
  Hardware is Lance, address is 0010.11d9.d001 (bia 0010.11d9.d001)
  Internet address is 23.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,

 (output cut for briefness)

FastEthernet0/1 is administratively down, line protocol is down
  Hardware is Lance, address is 0010.11d9.d002 (bia 0010.11d9.d002)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set

  (Output has been cut for briefness)

Serial0/2/0 is up, line protocol is up
  Hardware is HD64570
  Internet address is 24.0.0.1/8
  MTU 1500 bytes, BW 128 Kbit, DLY 20000 usec, rely 255/255, load 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  (Output has been cut for briefness)
```

(A) CDP

(B) PPP

(C) STP

(D) HDLC

449. Looking at the output in the figure below, what end of the point-to-point link is Serial0/2/0?

```
NY-R1#show controllers serial0/2/0
Interface Serial0/2/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 64000

(output has been cut for briefness)
```

(A) DCE

(B) DTE

(C) MPC8600

(D) PowerQUICC

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450. If an interface shows as administratively down, what command do you use to bring the interface up?

(A) `up`

(B) `enable`

(C) `no shutdown`

(D) `no down`

451. How do you view your ARP cache on a Cisco router? (Select two.)

(A) `show ip arp`

(B) `arp -a`

(C) `arp -show`

(D) `show arp`

452. Looking at the output in the figure below, what is the clock rate of serial0/2/0?

```
NY-R1#show controllers serial0/2/0
Interface Serial0/2/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 64000

(output has been cut for briefness)
```

(A) 35 Kbps

(B) 64 Kbps

(C) 128 Kbps

(D) 860 Kbps

453. You receive the following output from the `show interfaces` command. What should you check?

```
Serial0/2/0 is up, line protocol is down
```

(A) Check the physical aspects of the link, such as the cable.

(B) Check that the IP address is set.

(C) Check that the data link protocol is configured.

(D) Verify that a route is defined to the remote network.

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454. Looking at the output of the `show interfaces` command seen in the figure below, which of the following statements is true?

```
NY-R1>show interfaces

FastEthernet0/0 is up, line protocol is up
  Hardware is Lance, address is 0010.11d9.d001 (bia 0010.11d9.d001)
  Internet address is 23.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,

 (output cut for briefness)

FastEthernet0/1 is administratively down, line protocol is down
  Hardware is Lance, address is 0010.11d9.d002 (bia 0010.11d9.d002)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set

  (Output has been cut for briefness)

Serial0/2/0 is up, line protocol is up
  Hardware is HD64570
  Internet address is 24.0.0.1/8
  MTU 1500 bytes, BW 128 Kbit, DLY 20000 usec, rely 255/255, load 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  (Output has been cut for briefness)
```

(A) Serial port is using PPP.

(B) FastEthernet0/0 has no IP assigned.

(C) The bandwidth of FastEthernet0/0 is 1 Gbps.

(D) FastEthernet0/1 has not been configured.

455. You are having trouble communicating with a remote network. You have checked the status of the interfaces: Each interface is up, and the IP addresses look fine. What command do you use next?

(A) `show ip route`

(B) `show ip interface brief`

(C) `show controllers serial 0/0`

(D) `show cdp`

456. Which of the following commands would you use to troubleshoot issues at layer 3? (Select three.)

(A) `show ip interface brief`

(B) `show interfaces`

(C) `show protocols`

(D) `show controllers`

(E) `show ip route`

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457. What command do you use to view whether your serial port is the DCE or DTE device?

(A) `show interface serial 0/0`

(B) `show controllers serial 0/0`

(C) `show ip protocols serial 0/0`

(D) `show serial 0/0`

458. Which of the following commands would you use to check for issues at layers 1 and 2? (Select two.)

(A) `show ip interface brief`

(B) `show interfaces`

(C) `show protocols`

(D) `show controllers`

(E) `show ip route`

459. Which of the following could be an indication of excessive collisions or bad packets submitted by a malfunctioning network card? (Select two.)

(A) Runt

(B) Bandwidth

(C) CRC

(D) MTU

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460. Looking at the output in the figure below, how many packets have been received that were less than 64 bytes?

```
NY-R1>show interface fastethernet0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.63c6.0d01 (bia 0001.63c6.0d01)
  Internet address is 10.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 1150 bits/sec, 1 packets/sec
  5 minute output rate 1150 bits/sec, 1 packets/sec
     1689 packets input, 216192 bytes, 0 no buffer
     Received 4 broadcasts, 6 runts, 0 giants, 0 throttles
     0 input errors, 3 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     1693 packets output, 216308 bytes, 0 underruns
     0 output errors, 87 collisions, 1 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
```

(A) 3

(B) 6

(C) 87

(D) 216,380

461. You would like to see the number of runt packets received. What command would you use?

(A) `show cdp`

(B) `show cdp neighbors detail`

(C) `show ip interface brief`

(D) `show interfaces`

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462. You are troubleshooting performance issues on the network and suspect that a high number of packets are being retransmitted due to collisions. You use the show interface command to view the status on the interface. Looking at the output in the figure below, how many packets have been retransmitted due to collisions?

```
NY-R1>show interface fastethernet0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.63c6.0d01 (bia 0001.63c6.0d01)
  Internet address is 10.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 1150 bits/sec, 1 packets/sec
  5 minute output rate 1150 bits/sec, 1 packets/sec
     1689 packets input, 216192 bytes, 0 no buffer
     Received 4 broadcasts, 6 runts, 0 giants, 0 throttles
     0 input errors, 3 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     1693 packets output, 216308 bytes, 0 underruns
     0 output errors, 87 collisions, 1 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
```

(A) 3

(B) 6

(C) 87

(D) 216,380

Using Debug Commands

463–465 Choose the best answer(s).

463. Which of the following commands is recommended to disable debugging?

(A) `no debug ip all`

(B) `no debug ip nat`

(C) `no debug ip rip`

(D) `no debug all`

464. You are troubleshooting problems with RIP and suspect that your router is no longer receiving RIP updates. What command would you use to help diagnose the problem?

(A) `debug ip rip`

(B) `debug all`

(C) `no debug all`

(D) `show interfaces`

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465. You would like to view the details of all packets sent or received by the router. What command would you use?

(A) `debug ip rip`

(B) `debug ip route`

(C) `debug ip ospf`

(D) `debug ip packet`